Assignment 1 : Healthcare Risk Prediction — Machine Learning vs Neural Networks

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1. Problem Understanding

Diabetes is a serious chronic disease that affects millions globally and can lead to severe complications if not detected early. Predicting diabetes risk using patient health indicators enables early diagnosis and preventive care.

This project aims to build and compare:

- A **Logistic Regression model**
- A **Neural Network model**

Both models are trained to classify patients as diabetic or non-diabetic.

2. Data Exploration Insights

The dataset contains **6000 records and 7 features**, including one target variable.

Features in the Dataset

- `patient_id`: Unique identifier for each patient
- `age`: Age of the patient
- `bmi`: Body Mass Index
- `exercise_hours_per_week`: Weekly physical activity
- `smoker`: Smoking status (0 = No, 1 = Yes)
- `blood_pressure`: Blood pressure level
- `diabetes`: Target variable (0 = No, 1 = Yes)

All features are numerical, making the dataset suitable for machine learning models without additional encoding.

The distribution of the target variable (`diabetes`) is:

- **Non-diabetic (0): 5458 patients (~91%)**
- **Diabetic (1): 542 patients (~9%)**

This indicates a **class imbalance problem**, which affects model performance.

Key observations:

- BMI and blood pressure are strong predictors
- Lower exercise levels are associated with diabetes
- Age shows a positive relationship with diabetes risk

3. Machine Learning Model — Logistic Regression

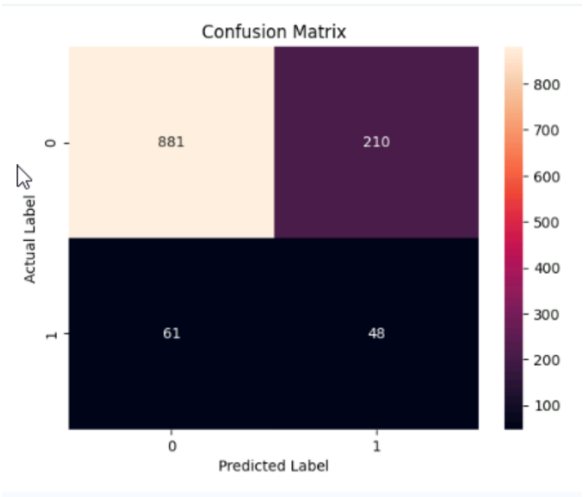
Logistic Regression was used as a baseline model.

Steps performed:

- Data split into training (80%) and testing (20%)
- Feature scaling applied using StandardScaler
- Class weights used to handle imbalance

Results:

- Accuracy: **77.42%**




```
Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.81      0.87      1091
           1       0.19      0.44      0.26       109

    accuracy                           0.77      1200
   macro avg       0.56      0.62      0.56      1200
weighted avg       0.87      0.77      0.81      1200
```

Key insight:

- Strong performance on non-diabetic cases

- Moderate ability to detect diabetes

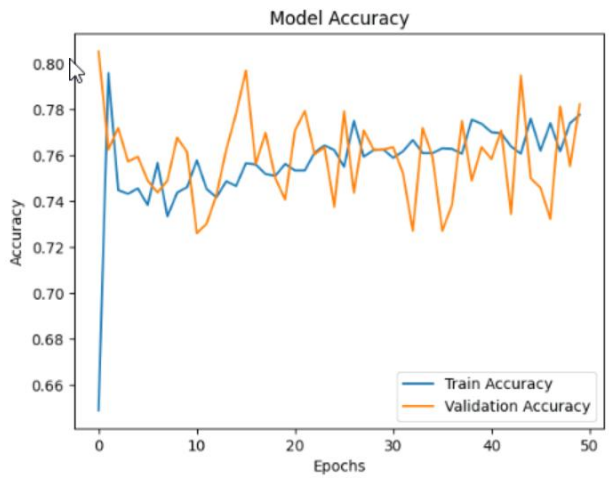
4. Neural Network Model

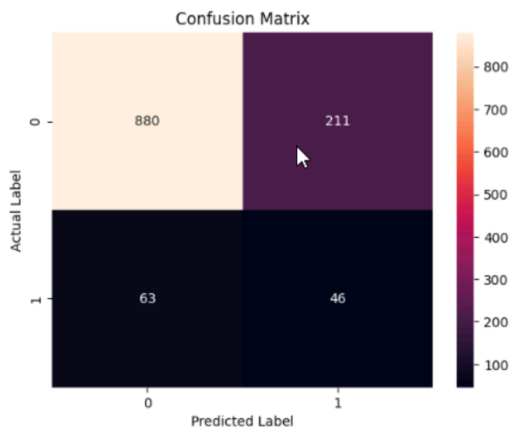
A Sequential Neural Network was built with three hidden layers.

Architecture:
• 32 → 16 → 8 neurons (ReLU activation)
• Output layer with Sigmoid activation

Training:
• 50 epochs
• Adam optimizer
• Class weights applied

Results:
• Accuracy: 77.17%





```
Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.81      0.87      1091
           1       0.18      0.42      0.25       109

    accuracy                           0.77      1200
   macro avg       0.56      0.61      0.56      1200
weighted avg       0.86      0.77      0.81      1200
```

Key insight:
• Comparable performance in detecting diabetes cases
• Slightly lower overall accuracy than Logistic Regression

5. Model Comparison & Analysis

5.1 Accuracy Comparison

• Logistic Regression Accuracy: 77.42%
• Neural Network Accuracy: 77.17%

The Logistic Regression model achieved slightly higher overall accuracy compared to the Neural Network.

5.2 Performance on Diabetic Class

Metric	Logistic Regression	Neural Network
Precision	0.19	0.18
Recall	0.44	0.42
F1 Score	0.26	0.25

Key Insight
• Logistic Regression has slightly higher recall (0.44 vs 0.42)
• Logistic Regression also has slightly better precision

This means:
• Logistic Regression detects more actual diabetes cases
• Neural Network performs similarly but with slightly lower effectiveness

5.3 Why Neural Network Did Not Outperform

Although Neural Networks are powerful, in this case:
• The dataset is structured and relatively small (6000 rows)
• Relationships between features may be mostly linear
• Logistic Regression is well-suited for such data

5.7 When to Use Each Model

Use Logistic Regression when:

• Interpretability is important (e.g., healthcare decisions)
• Dataset is small to medium-sized
• Relationships are mostly linear

Use Neural Networks when:

• Dataset is large and complex
• Non-linear relationships are present
• Accuracy improvement is critical

6. Conclusion

Both models successfully predicted diabetes risk.

• Logistic Regression is more reliable for this dataset
• Neural Network shows potential for complex data